

SILTHAUL · DESIGN DECISIONS REGISTER

SiltHaul design decisions register

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DOCUMENT CONTROL

Document ID	SLH-DEC-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened; every decision made under Amish's pre-approval of 2026-10-03

SiltHaul design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below set SiltHaul's safety case (overload limit, holding, anchors, dumping). Each took the conservative option; the evidence that would relax it is in SLH-DDR-001, Table 1.

■ Open decisions

None. All decisions were made under Amish's 2026-10-03 pre-approval.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Release load of the shear pins from the bar actually bought	The 4 mm pin is sized on 240 MPa shear; a stronger bar raises the overload limit above 3,000 N	SLH-CAL-001, F; R11
2	Recommended loads of the M12 wedge anchor bought, in the concrete grade expected	The tail block check uses 4 kN tension and 6 kN shear	SLH-CAL-001, I; R7
3	Rope breaking strength and splice efficiency of the rope bought	Factor 5.4 at the limit assumes 18 kN and 90 % at the splice	SLH-CAL-001, G; R6
4	Sheave bore and rating	Pins are 25 mm; the sheaves must be rated at least 1,000 kg	BOM line 14
5	Bearing centre heights and bolt pitches	The pads and top plates are drilled to UCP206 and UCP205 catalogue sizes	SLH-DWG-101
6	Sprocket bores and the 12-tooth sprocket's fit on the 36 mm hub	Sets the hub and the line-up with the 48-tooth wheel	SLH-DWG-104

■ Value engineering

Value-engineering target: USD 2,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 887.50 (USD 1,112.50 under the target). Main cost drivers and savings worth trying:

- The winding drum (USD 98) and the box (USD 78) are the largest made parts; the ramp (USD 73) and the rope (USD 71.50) follow.
- Bought lifting gear (sling, shackles, bridles, USD 110) is kept at rated, traceable grades because it carries the safety case.
- Savings worth trying: a drum from a scrap gas cylinder or pipe offcut of the same diameter; a timber ramp with a steel wear strip; buying rope with factory eye splices in bulk.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	Split winding drum in place of a friction capstan; 4:1 chain drive with two cranks	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, items 1, 2
2026-10-03	Shear pin limits rope tension to about 2.5 kN; all parts sized on 3,000 N; requirement R11 added	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 3
2026-10-03	Two opposite ratchet wheels and pawls on the drum shaft; requirement R12 added	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 4
2026-10-03	Tail block on four drilled floor anchors only, never walls or door frames	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 5
2026-10-03	Capstan anchored by a round sling to a tree or vehicle, level or rising 10 degrees at most; stakes only against skating	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 6
2026-10-03	10 mm polyester double braid, 18 kN	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 7
2026-10-03	Box open toward the door, sloped back, forward tipping; two people on the tipping bar	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, items 8, 9
2026-10-03	Capstan at least 6 m beyond the door	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 10
2026-10-03	First co-design candidate to approach: Kerala State Disaster Management Authority (not agreed)	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 11
2026-10-03	SiltHaul capstan recorded as the candidate common block for SaltDrag; CalRig as the first candidate proof-load rig	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, item 12
2026-10-03	budget_usd kept at 2,000 as a value-engineering target	Amish: "I also accept any cost overruns or variations from the assumed scope cost."	SLH-DDR-001, item 13
2026-10-03	R1 kept as the trial target and reported not met on paper	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-001, Consequences
2026-10-03	Design for construction: the thirteen changes of SLH-DDR-002	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	SLH-DDR-002
2026-10-03			docs/REVIEW.md, TRL 3

Date	Decision	Decided by	Record
	Appearance model additions for renders: wound rope on the drum, stakes above ground only, context wall, tree and mannequin	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	